Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

– Can Tho, May 2026 –

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I. Project Report

1. Status Report

#	Work Item	Status	Notes (Work Item in Details)
1		Pending	
2		In Progress	
3		Completed	

2. Team Involvements

#	Task	Member	Notes (Task Details, etc.)
1		KienNT	
2		TuanTV	
3		AnhLM	

3. Issues/Suggestions

#	Issue	Status	Notes (Solution, Suggestion, etc.)
1		Pending	
2		In Progress	
3		Completed	

II. Project Management Plan

1. Overview

1.1 WBS & Estimation

#	WBS Item	Complexity	Est. Effort (man-days)
1	***Document***		***45***
1.1	Report 1: Project Introduction	Simple	4
1.2	Report 2: Project Management Plan	Medium	7
1.3	Report 3: Software Requirement Specification (SRS)	Complex	10
	Report 4: Software Design Document (SDD)	Complex	12
	Report 5: Test Documentation	Medium	7
	Report 6: Software User Guides	Simple	2
	Report 7: Final Project Report	Simple	3
2	***Design***		***25***
2.1	Design Prototype	Medium	10
2.2	Design database	Medium	5
2.3	Design AI workflow	Medium	5
2.4	Design AI network architectures	Medium	5
3	**Implement**		**480**
3.1	Register	Medium	4
3.2	Login	Medium	4
3.3	Logout	Medium	3
3.4	Login with Google	Medium	9
3.5	Reset password	Medium	8
3.6	Forget password	Medium	4
3.7	View profile	Medium	3
3.8	Update profile	Medium	4
3.9	Send OTP	Medium	4
3.10	View roles	Medium	3
3.11	View permissions	Medium	4
3.12	Create role	Medium	9

3.13	Update role permission	Medium	4
3.14	Assign Role	Medium	4
3.15	Remove role	Medium	4
3.16	Access log	Medium	4
3.17	Create digital signature	Medium	3
3.18	View digital signature	Medium	4
3.19	Delete digital signature	Medium	3
3.20	Verify Identity KYC	Medium	8
3.21	Update digital signature	Medium	4
3.22	Ban user	Medium	4
3.23	Unban user	Medium	3
3.24	View clinics list	Medium	3
3.25	View clinic details	Medium	4
3.26	Update clinic information	Medium	4
3.27	Add treatment room	Medium	3
3.28	View treatment rooms	Medium	4
3.29	Update treatment room	Medium	3
3.30	Delete treatment room	Medium	9
3.31	Create working schedule	Medium	4
3.32	Manage working schedule	Medium	3
3.33	View personal schedule	Medium	8
3.34	Request personal schedule change	Medium	4
3.35	Update personal schedule	Medium	4
3.36	Notify Schedule Change	Medium	4
3.37	Notify Shift Transfer	Medium	4
3.38	Add patient profile	Medium	3
3.39	View patient profile	Medium	4
3.40	Update patient profile	Medium	4
3.41	Add medical record	Medium	3
3.42	View medical record	Medium	4

3.43	Update medical record	Medium	3
3.44	Export medical record	Medium	7
3.45	Add treatment record	Medium	3
3.46	View treatment record	Medium	7
3.47	Update treatment record	Medium	4
3.48	Export treatment record	Medium	8
3.49	Create appointment at Facility	Medium	12
3.50	Create appointment at Specialty	Medium	10
3.51	Create appointment with Doctor	Medium	10
3.52	Create after-hours appointment	Medium	3
3.53	View appointments	Medium	4
3.54	Update appointment	Medium	3
3.55	Cancel appointment	Medium	4
3.56	Confirm appointment	Medium	4
3.57	View payment history	Medium	4
3.58	Initiate payment	Medium	12
3.59	Refund payment	Medium	8
3.60	Chatbot Support for Booking	Medium	8
3.61	View specialties	Medium	20
3.62	Add specialty	Medium	3
3.63	Update specialty	Medium	3
3.64	Delete specialty	Medium	4
3.65	Input symptoms	Medium	3
3.66	View symptoms	Medium	4
3.67	Update symptoms	Medium	3
3.68	Add Clinical Notes	Medium	3
3.69	Delete symptoms	Medium	3
3.70	Create treatment plan	Medium	15
3.71	View treatment plan	Medium	3
3.72	Update treatment plan	Medium	6

3.73	Approve treatment plan	Medium	3
3.74	Delete treatment plan	Medium	3
3.75	Issue digital prescription	Medium	8
3.76	Order X-ray / CBCT imaging	Medium	10
3.77	Order laboratory services	Medium	8
3.78	Order clinical tests	Medium	6
3.79	Upload endodontic images	Medium	4
3.80	Upload X-ray / CBCT images	Medium	4
3.81	View dental image library	Medium	3
3.82	Attach images to treatment record	Medium	8
3.83	Update images in treatment record	Medium	6
3.84	View doctor performance report	Medium	16
3.85	View doctor dashboard	Medium	10
3.86	View customer dashboard	Medium	10
3.87	View financial report	Medium	14
		Total Estimated Effort (man-days)	***550***

1.2 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	AI Model Performance – AI recommendations might be inaccurate, affecting user trust.	High	Medium	Regularly test and fine-tune AI models.
2	Integration Challenges - Issues in integrating e-wallets or payment systems (VNPay, PayOS).	Medium	Sometimes	Collaborate with service providers during integration.
3	Member withdraws from project	High	Low	Assign work flexibly from the beginning; always have backup for each part. When a member withdraws, the team leader reviews the progress and re-assigns appropriately to ensure deadlines.
4	Source code missing	High	Low	Use Git to backup code regularly, clearly divide tasks for each person.

2. Management Approach

2.1 Project Process

We have selected the Agile Software Development Model due to its iterative and incremental framework. By decomposing the development process into brief cycles, Agile facilitates continuous product delivery while fostering rigorous collaboration between developers and stakeholders. This approach prioritizes dynamic responsiveness to evolving requirements, ensuring the final output maintains high utility and alignment with contemporary market demands

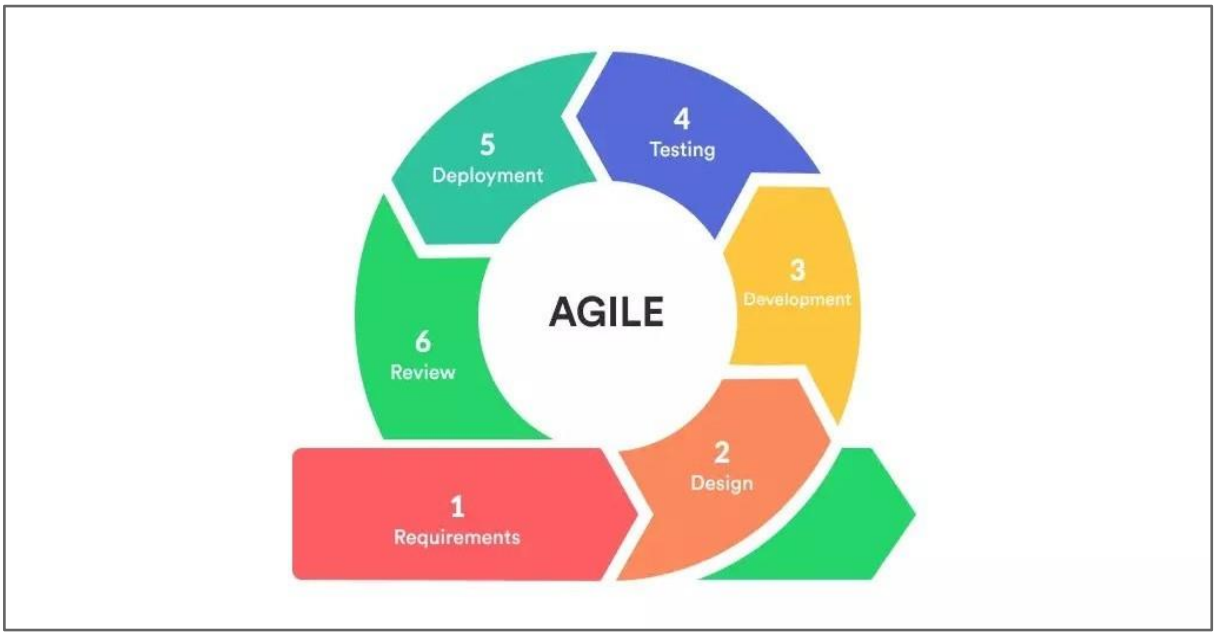

Figure 1. Agile development cycle[1]

a. *Advantages of Agile:*

Agile allows for fast and frequent software delivery, allowing customers to experience and provide feedback to improve the product early. The flexibility of this method helps the team easily adapt to changing requirements, minimizing risks through integrating early feedback. Continuous testing and integration ensure high software quality, minimizing deviations during the development process. Agile also facilitates rapid testing and adjustment, keeping the project on track.

b. *Disadvantages of Agile:*

Despite its many benefits, Agile requires strong communication skills, high commitment, and strong discipline from teams to ensure efficiency in short development cycles. The lack of detailed planning up front can lead to scope creep if not managed carefully. Additionally, Agile may not be suitable for projects with fixed requirements or strict compliance requirements.

c. *Why we used Agile:*

The project chose Agile to ensure rapid delivery of core features while remaining adaptable to changes. Agile allows the team to respond quickly and maintain high quality through continuous testing and integration cycles. Ensuring that the final product not only meets expectations but also delivers optimal value makes Agile the ideal choice for this project.

[1] https://slite.com/learn/agile

2.2 Quality Management

Maintaining superior product standards is a fundamental objective in the engineering of the SMILE platform. To ensure a robust and dependable architecture, the project adopts a dual-strategy approach combining proactive mitigation with reactive verification fully integrated into the Agile Scrum lifecycle.

The framework is structured around four critical technical practices:

- Defect Prevention: Early-stage risk reduction is achieved through rigorous requirement synthesis and adherence to standardized coding protocols. This minimizes technical debt and stabilizes the core infrastructure.
- Code Reviewing: Systematic peer evaluations are mandated to uphold architectural integrity and facilitate collective knowledge transfer. These sessions act as high-level checkpoints for identifying latent inefficiencies.
- Integration Testing: Regular execution of inter-component tests validates seamless data exchange across modules, surfacing interface discrepancies that isolated unit tests might overlook.
- System Testing: A holistic validation process evaluates the platform's end-to-end functionality, security posture, and cross-environment compatibility to ensure all technical benchmarks are satisfied before release.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
NextJS	All Members	Week 1 – 2 sessions, 2h each	Mandatory
NestJS	All Members	Week 1 – 3 sessions, 1.5h each	Mandatory
Git, Github	All Members	Week 2 – 2 sessions, 2h each	Mandatory

3. Master Schedule

#	Deliverable	Due Date	Deliverable Scope
1	Report 1	18/05/2026	Introduce project, existed project, business opportunity, vision, scope and limit of project
2	Report 2	21/05/2026	Plan of project, software development model, manage quality of project, configuration management, team and role, responsibility in project, training plan and communication schedule during development project.

3	Report 3	14/06/2026	User requirement, functional requirement, requirement specification, business rules
4	Report 4	14/07/2026	Architecture Design, Detailed Design, Database Design
5	Report 5	19/07/2026	Test model, testing level, test plan, test case and test report
6	Report 6	29/07/2026	Release package, installation guides, user manual
7	Report 7	02/08/2026	Project introduction, project management plan, software requirement, software design, software testing, user guides
8	Final Thesis	05/08/2026	Project introduction, project management plan, software requirement, software design, software testing, user guides
9	Code backend	05/08/2026	Code & Unit test, Integration test cases
10	Code frontend	05/08/2026	Code & Unit test, Integration test cases
11	Database	27/05/2026	Design database

4. Project Organization

4.1 Team & Structures

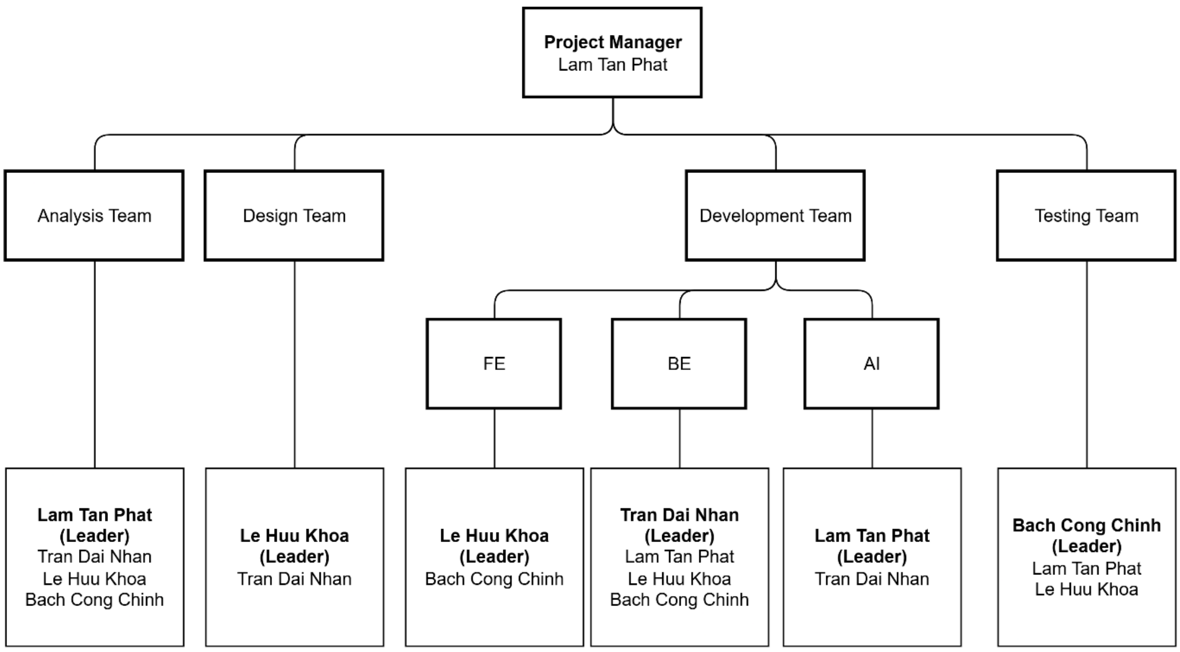

Figure 2. Team Structure

4.2 Roles & Responsibilities

Role	Responsibility

Project Manager	Responsible for planning, executing, and delivering the project while managing resources, risks, and stakeholders to meet project goals successfully.
Analysis Leader	Responsible for gathering, analysing, and documenting business requirements, translating them into technical specifications.
Analysis Member	Assist in gathering and documenting requirements, conducting research, and providing valuable insights to support the project's analysis activities.
Design Leader	Guides the visual and UI/UX design of the project, collaborates with stakeholders, and oversees the creation of high-quality visual concepts and user interfaces.
Design Member	Assists in creating visual concepts and user interface designs, collaborating with the design team and contributing to the overall design process.
Development Leader	Oversees the technical aspects of the project, guiding the development team to deliver high-quality software solutions.
Development Member	Responsible for coding, implementing, and testing software solutions in accordance with project requirements, collaborating with the Development Leader and other members.
Test Leader	Leads the testing activities, develops test strategies, manages the testing team and ensures software quality through effective testing processes.
Test Member	Executes test cases, identifies and reports defects, and contributes to the overall quality of the software.

5. Project Communication

5.1 Communication Plan

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Daily Scrum	Lam Tan Phat Tran Dai Nhan Le Huu Khoa Bach Cong Chinh	Synchronize our activities, discuss progress, and plan our work for the next 24 hours	Every day 22h	Google Meet, Discord
Sprint Review	Lam Tan Phat Tran Dai Nhan Le Huu Khoa Bach Cong Chinh	Showcase the features, gather feedback, refine the backlog, and improve the product and team processes	Every weekend 22h	Offline and Google Meet

5.2 External Interface

a. FU Contacts

Function	Contact Person (name, position)	Contact address (email, telephone)	Responsibility

Supervisor	Luong Hoang Huong	HuongLH3@fe.edu.vn 0941666545	- Provide document template - Give instruction to project team - Review deliverables - Supervise project status and process

b. Customer Contacts

Function	Contact Person (name, position)	Contact address (email, telephone)	Responsibility
Supervisor	Luong Hoang Huong	HuongLH3@fe.edu.vn 0941666545	- Provide requirement about project - Confirm requirements, approve documents, design and project plan - Participate in meetings with project team

6. Configuration Management

6.1 Tools & Infrastructures

Programming languages	TypeScript, JavaScript, Python
Framework	NestJS, ReactJS, NextJS,
API	RESTful API
DBMS	PostgreSQL, Redis
IDEs/Editors	Visual Studio Code
UML tools	PlantUML, draw.io, dbdiagram.io
Version Control	Git/Github
Deployment server	Docker and Server: https://smile-dental.vn/
Project management tool	Excel

6.2 Document Management

In the SMILE project, our team prioritizes using Microsoft Word and Microsoft Excel to manage project documents.

6.2.1 Microsoft Word

6.2.2.1 Introduction

Microsoft Word is a sophisticated word-processing application developed by Microsoft. It facilitates the creation, formatting, and sharing of professional documents. Integrated within the Microsoft 365 ecosystem, it supports both local desktop editing and real-time cloud-based collaboration via web interfaces.

a. Characteristics:

- Advanced Document Composition: Provides a comprehensive suite of tools for text formatting, page layout design, and graphical integration.
- Co-authoring & Collaborative Editing: Enables multiple stakeholders to review and edit documents simultaneously. Revision history and "Track Changes" ensure auditability and precision during the drafting process.
- Granular Security: Features robust permission management, allowing administrators to restrict access or encrypt sensitive documents to maintain project confidentiality

Figure 3. Microsoft Word[2]

b. Advantages:

- Accessibility: Offers a seamless transition between offline desktop software and the online web version.
- Industry-Standard Tools: Equipped with advanced citation management, automated indexing, and a vast library of templates.
- Synchronous Updates: Edits are reflected in real-time across all connected devices when stored on OneDrive or SharePoint.
- Cross-Platform Integration: High compatibility with other Microsoft Office applications (Excel, PowerPoint) for data embedding.

c. Why we used:

- Document Versioning: Using Word to manipulate and secure document versions provides a continuous auto-save safety net, preventing data loss during drafting.
- Teamwork Workspace: Creating a dynamic workspace for team editing allows members to draft concurrently, speeding up the process without creating conflicting files.
- Smooth Workflow: Leveraging familiar Microsoft tools keeps the workflow efficient, skipping steep learning curves so the team can focus entirely on getting the work done.

6.2.2 Microsoft Excel

6.2.2.1 Introduction

Microsoft Excel is a powerful spreadsheet application developed by Microsoft, designed for data organization, complex calculations, and professional analysis. As a core component of the Microsoft 365 suite, it enables users to manage large datasets through both a robust desktop client and a synchronized web-based platform.

a. Characteristics:

- Advanced Data Processing: Utilizes a grid-based system for structured data entry, supporting sophisticated mathematical models and logical algorithms.

[2] https://word.cloud.microsoft/en-us/

- Real-time Collaboration: Facilitates simultaneous multi-user access via OneDrive or SharePoint. Administrators can utilize "Version History" to audit modifications and ensure data integrity.
- Security & Permissions: Employs granular access controls, allowing owners to protect specific cells, sheets, or entire workbooks with encryption and restricted editing rights.

Figure 4. Microsoft Excel[3]

b. Advantages:

- Extensive Functional Library: Supports an exhaustive range of formulas and functions for statistical, financial, and engineering calculations.
- Hybrid Connectivity: Operates seamlessly in both offline environments and online cloud modes, ensuring constant accessibility.
- Data Visualization: Includes advanced charting tools and Pivot Tables to transform raw data into actionable insights.
- Automation Capabilities: Supports macros and VBA (Visual Basic for Applications) to streamline repetitive workflows.

c. Why we used:

- Data Reliability: Automatic cloud-saving features prevent data loss during complex analysis phases.
- Collaborative Synergy: Enables multiple team members to update financial or technical datasets concurrently without version conflicts.
- Expertise Alignment: Utilizes the team's existing technical proficiency in Excel to maximize analytical efficiency and minimize the learning curve.

6.3 Source Code Management

To manage the project source code, our team uses Git, Github for easier management.

6.3.1 Git

6.3.1.1 Introduction

Git is a widely used distributed version control system for tracking changes to source code during software development.

[3] https://excel.cloud.microsoft/en-us/

Figure 5. Git[4]

a. Characteristics:

- Distributed Version Control: full source code storage with complete history and full version tracking.
- Branches: Allows users to create multiple branches for different features or versions. Each branch can be worked on independently.
- Merge: Facilitates merging of branches. Git handles merges efficiently and can automatically resolve many conflicts.

Figure 6. Github[5]

b. Advantages:

- Git is completely free and open source.
- Save versions of software source code.
- Allow offline work.
- Restore source code if needed.

c. Why we used:

- Work online in real time with multiple people.
- Convenient for control and security.
- Team members have been trained for a long time.

6.3.1.2 Management mechanism

The leader creates a common branch called "main". Each member creates a branch with code according to the git structure. Each time using the code, pull it back and edit it on their own branch. Then push it to the master branch and wait for Leader to approve it before it can be merged.

6.3.2 Github

6.3.2.1 Introduction

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[4] https://git-scm.com/
[5] https://github.com/

GitHub is a project management and code versioning system. Developers can clone source code from a repository.

a. Characteristics:

- Source code management: Github allows users to create a repo, the entire source code of that repo is stored on GitHub. And multiple accounts can operate on a repo, making project management easier.
- Tracking version: Users can track the changes that users have pushed to the repo.
- Collaboration and Community: Developers can interact with each other on Github in a repo, track bugs, fix them and push them to the repo. Ensure the product is always at the best level.

b. Advantages:

- Is a free cloud web platform.
- Source code storage.
- Allows multiple people to work on a project.
- Pull and push request management features make it easier to submit and push code.
- Easy-to-use web interface.
- Track source code changes.

c. Why we used:

- Securely store source code on the cloud.
- Pull and push code online fast in real time.
- The application has been thoroughly trained.

6.3.2.2 Management mechanism

The leader creates a repository containing the project. It is the interface for managing the branches that have been pushed. Every time a member pushes from their own branch, they must create a pull request to merge the code to the master branch to synchronize the code.